Problem Statement 1: Design a LEX Code to count the number of lines, space, tab-meta character and rest of characters in a given input pattern.

Code:-

```
%{
#include<stdio.h>
int lines=0, spaces=0, tab=0, ch=0;
%}

%%
" "     {spaces++;}
"\t"    {tab++;}
"\n"    {lines++;}
.       {ch++;}
%%

int yywrap(){
   return 1;
}

int main(){
   printf("Enter a string :\n");
   yylex();
   printf("No. of lines:%d\n", lines);
   printf("No. of tabs:%d\n", tab);
   printf("No. of spaces:%d\n", spaces);
   printf("No. of characters:%d\n", ch);
   return 0;
}
```

Output:-

```
achintyabist@Achintyas-MacBook-Air compiler_design % lex Q1.l
achintyabist@Achintyas-MacBook-Air compiler_design % gcc lex.yy.c
achintyabist@Achintyas-MacBook-Air compiler_design % ./a.out
Enter a string :
Hello, How are you.
I       am      fine.
No. of lines:2
No. of tabs:2
No. of spaces:3
No. of characters:24
```

Problem Statement 2: Design a LEX Code to count and print the total number of characters, words, white spaces and lines in a given file named as 'Input.txt'.

Code:-

```
%{
#include <stdio.h>
int newlines=0,tabs=0,whitespaces=0,characters=0;
%}
%%
\n          {newlines++;}
"\t"        {tabs++;}
" "         {whitespaces++;}
.           {characters++;}
%%
int yywrap()
{
  return 1;
}
int main()
{
  extern FILE *yyin;
  yyin=fopen("input.txt","r");
  yylex();
  printf("No. of newlines : %d\n",newlines);
  printf("No. of tabs : %d\n",tabs);
  printf("No. of whitespaces : %d\n",whitespaces);
  printf("No. of characters : %d\n",characters);
};
```

Input:-

input.txt:

input.txt

```
Hello, How are you.
I   am  fine.
```

Output:-

```
achintyabist@Achintyas-MacBook-Air compiler_design % lex Q2.l
achintyabist@Achintyas-MacBook-Air compiler_design % gcc lex.yy.c
achintyabist@Achintyas-MacBook-Air compiler_design % ./a.out
No. of newlines : 2
No. of tabs : 2
No. of whitespaces : 3
No. of characters : 24
```

Problem Statement 3: Design a LEX Code to replace all the white spaces of 'Input.txt' file by a single blank character and store the output in 'Output.txt' file.

Code:-

```
%{
#include <stdio.h>
%}

%%
[ \t\n]+    {fprintf(yyout, " ");}
.       {fprintf(yyout, "%s", yytext);}
%%

int yywrap()
{
   return 1;
}

int main()
{
   extern FILE *yyin, *yyout;
   yyin  = fopen("input.txt", "r");
   yyout = fopen("output.txt", "w");
   yylex();
   return 0;
}
```

Input:-

input.txt:

input.txt

```
Hello,     How
    are you   .
I   am      fine.
```

Output:-

```
achintyabist@Achintyas-MacBook-Air compiler_design % lex Q8.l
achintyabist@Achintyas-MacBook-Air compiler_design % gcc lex.yy.c
achintyabist@Achintyas-MacBook-Air compiler_design % ./a.out
```

output.txt:

output.txt

```
Hello, How are you . I am fine.
```

Problem Statement 4: Design a LEX Code to remove the comments from any C-Program (in.c) given at runtime and store into 'out.c' file.

Code:-

```
%{
#include <stdio.h>
FILE *yyin;
FILE *out;
%}

%%
"//".*                          ;
"/*"([^*]|\*+[^*/])*"*"+"/"        ;
.|\n                            { fputc(yytext[0], out); }
%%

int yywrap() {
   return 1;
}

int main() {
   yyin = fopen("input.c", "r");
   out  = fopen("output.c", "w");
   if (yyin == NULL || out == NULL) {
      printf("Error opening files\n");
      return 1;
   }
   yylex();
   fclose(yyin);
   fclose(out);
   return 0;
}
```

Input:-

input.c:

```c
#include <stdio.h>   // standard input output header
/*
   This is a multi-line comment
   explaining the main function
*/
int main() {
    int a = 10;      // variable declaration
    int b = 20;      // another variable
    /* multi-line comment inside function
       calculating sum of two numbers */
    int sum = a + b;
    printf("Sum = %d\n", sum); // print the result
    return 0;   // end of program
}
```

Ouput:-

output.c:

```c
C output.c > main()
#include <stdio.h>

int main() {
    int a = 10;
    int b = 20;

    int sum = a + b;
    printf("Sum = %d\n", sum);
    return 0;
}
```